

Solution Requirements

Date	15 March 2026
Team ID	xxxxxx
Project Name	Rising Water
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users must securely log in or verify their identity. Example: Implement multi-factor authentication for all users.	High
2	Authorization levels	Define user roles and specify what access or permissions each role has. Example: Admins can manage users, while regular users can only view their own data.	medium
3	External interfaces	<i>Detail any third-party APIs, external databases, or services the system must connect to. Example: Integrate with a</i>	medium

		weather data API for real-time flood alerts.	
4	Transactions processing	Explain how data exchanges, payments, or core system transactions will be handled. Example: Securely process user-submitted flood reports and store them in the database.	high
5	Reporting	Describe the types of reports, dashboards, or data exports the system must generate. Example: Generate monthly flood risk assessment reports for stakeholders.	high
6	Business rules, etc.	List specific logic, constraints, or workflows unique to the business process. Example: Automatically escalate high-risk flood alerts to emergency services.	low
7	Compliance to laws or regulations	Specify any legal, industry, or data privacy standards the system must adhere to (e.g., GDPR, HIPAA). Example: Ensure all user data is stored in compliance with GDPR.	low
8	<i>Other</i>	Add any additional functional requirements as needed. Example: Provide a mobile-friendly interface for field agents.	low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	The system must deliver fast response times and efficient processing	Example Target: Page loads in less than 2 seconds.
2	Scalability	The solution should handle growth in users and data volume without degradation in performance.	Example Target: Supports up to 10,000 concurrent users
3	Security & Data Privacy	Example Target: Supports up to 10,000 concurrent users	Example Target: End-to-end AES-256 encryption for all sensitive data.
4	Reliability & Availability	The system should be highly available and fault-tolerant to minimize downtime.	Example Target: 99.9% uptime per month.
5	Usability & Accessibility	The user experience must meet accessibility standards and be intuitive for all users.	Example Target: Complies with WCAG 2.1 AA standards
6	<i>Other</i>	Additional requirements may include ease of maintenance and the ability to operate across different platforms	